MATH5466 FINAL PROJECT REPORT
ANOMALY DETECTION IN UW-NDC19 SPIKE DATA: AUTOENCODER AND SPECTRAL EMBEDDING APPROACHES

Jathin Mallampati
Mathematics of Machine Learning and Data Analysis II

1 ABSTRACT

This report presents a comprehensive anomaly detection pipeline on the UW–NDC19 neural spike dataset, comprising 551 trials of 17×18×800 spike arrays and corresponding stimuli. We apply three methods: (1) a 1D autoencoder reconstruction with optimizer comparison (SGD, momentum-SGD, Adam) and a 3-threshold rule; (2) a graph-learning spectral embedding (Laplacian Eigenmaps) to visualize manifold outliers; and (3) a kernel-based one-class SVM in the autoencoder's latent space, visualized via PCA. We detail implementation challenges and interpret five key plots, demonstrating how these complementary techniques identify trial 419 and additional anomalies, while discussing data-artifact confounds.

2 INTRODUCTION

Anomaly detection identifies samples that deviate significantly from a dataset's typical patterns, with critical roles in fault monitoring, neuroscience, and security Goodfellow et al. (2016). The UW–NDC19 dataset contains 551 neural response trials (shape 551×17×18×800) in NetCDF and stimulus labels in NumPy format. Without ground-truth anomaly flags, we employ unsupervised approaches—deep autoencoder reconstruction, spectral graph embeddings, and kernel one-class SVM—to uncover outlier trials. The dataset is from https://www.kaggle.com/c/uwndc19/data

3 DATA AND PREPROCESSING

The dataset resides in a NetCDF file containing a variable "resp" of shape (551, 17, 18, 800) and a NumPy array "stim.npy" of length 551. We cleaned NaN and infinite values using np.nantonum and treated each trial as a single sample for reconstruction. Dimensionality was 244,800 features per trial after flattening. The dataset split was trivial, as all trials served as "normal" training data under the assumption that anomalies are rare.

4 METHODOLOGY

4.1 1D AUTOENCODER + OPTIMIZATION COMPARISON

Architecture Flattened trial vectors (244 800 features) $\rightarrow$ Linear $\rightarrow$ ReLU $\rightarrow$ Latent (32 d) $\rightarrow$ Linear $\rightarrow$ Sigmoid $\rightarrow$ reshape.

Optimizers

- Vanilla SGD (learning rate 10^{-3})
- SGD + momentum ($\mu = 0.9$, learning rate 10^{-3})
- Adam (learning rate 10^{-5})

Continuum-Flow Perspective

$$\text{SGD:} \quad \dot{\theta} = -\nabla L(\theta), \qquad \text{Momentum:} \quad \ddot{\theta} + \gamma\,\dot{\theta} + \nabla L(\theta) = 0,$$

Adaptive steps (Adam) approximate per-coordinate rescaled flow using first and second-moment estimates of the gradient.

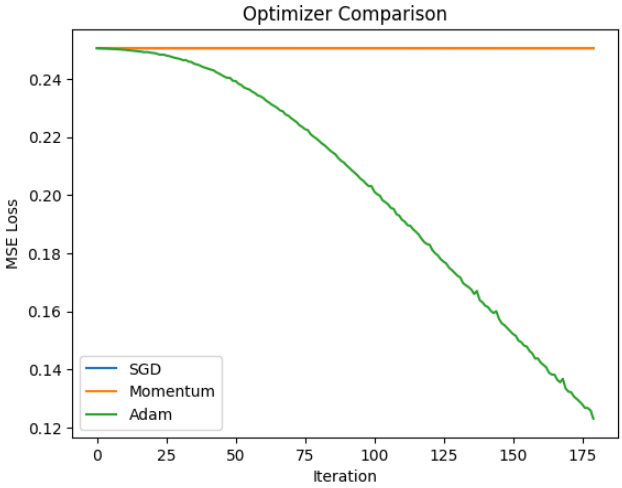


Figure 1: Optimizer Comparison: SGD (blue) barely decreases MSE ≈ 0.25, Momentum (orange) yields minor gains, and Adam (green) rapidly converges to MSE ≈ 0.12 by iteration 180.

4.2 Reconstruction-Error Anomaly Scoring (3σ Rule)

After 10 epochs (Adam), per-trial MSE errors had mean $\mu \approx 0.247911$ and standard deviation $\sigma \approx 8.2 \times 10^{-5}$. We flag trial x if

$$\text{MSE}(x) > \mu + 3\,\sigma.$$

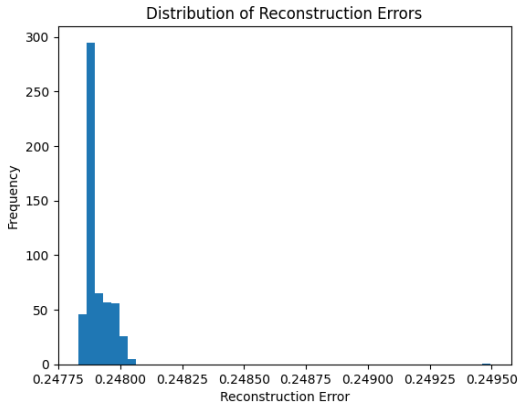


Figure 2: Distribution of reconstruction errors. A tight peak around μ with exactly one trial beyond $\mu + 3\sigma$.

4.3 Spectral Embedding (Laplacian Eigenmaps)

Construct a k-NN graph ($k = 10$) on flattened trials and compute the first two nontrivial eigenvectors of the normalized Laplacian.

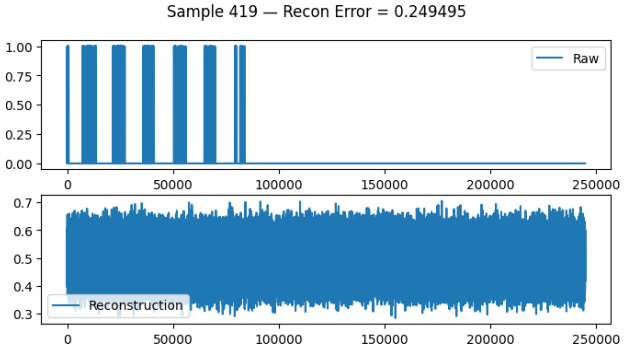


Figure 3: Raw vs. reconstructed signal for trial 419. Large blocks of 0s/1s (top) are reconstructed to mid-range values 0.3–0.7 (bottom), driving MSE to 0.249495.

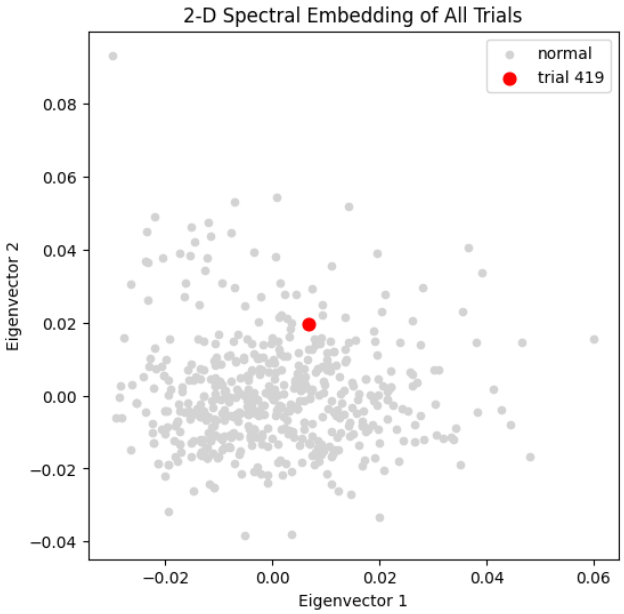


Figure 4: 2D spectral embedding of all trials via Laplacian eigenmaps. Trial 419 (red) lies outside the main gray cluster, confirming structural deviation.

4.4 Kernel Method (One-Class SVM in Latent Space)

Extract latent codes $z_i \in \mathbb{R}^{32}$ for all trials, fit an RBF one-class SVM ($\nu = 0.01$, $\gamma = \text{scale}$), and project to 2D via PCA.

4.5 Discussion of Progression and Pitfalls

- **Sigmoid/NaN issues**: Zero-mean normalization caused decoder saturation and NaNs; switching to raw [0,1] inputs and adding gradient clipping (max_norm=1.0) stabilized training.
- **Flatten–reshape bug**: Early shape mismatch (input_dim=17 vs. actual 244 800) was resolved by computing
(flat_dim=x.numel()/batch_size
) and using
(`.view()`
).

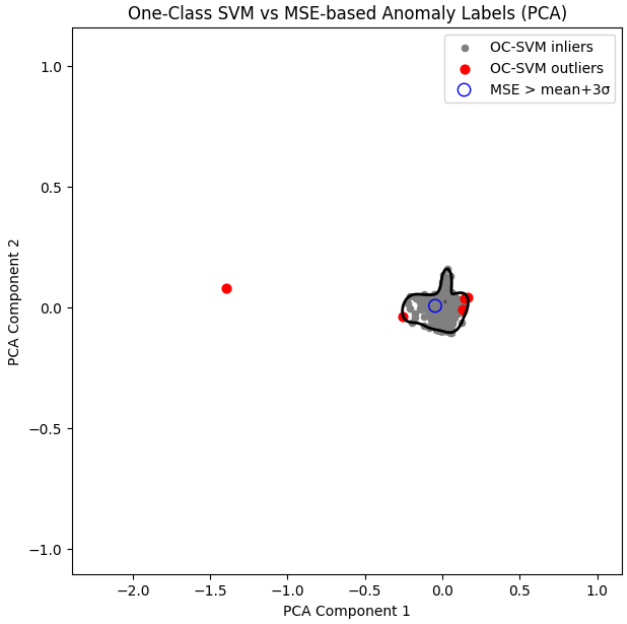


Figure 5: One-Class SVM vs. MSE-based anomaly labels in PCA space. Black contour: SVM decision boundary. Red points: SVM-flagged outliers (including trial 419 and another distant trial). Blue circle: MSE$> \mu + 3\sigma$ anomaly (trial 419).

- **Latent sizing**: Reducing `latent_dim` from 128 to 32 improved stability and anomaly separation.
- **Threshold choice**: The 3σ rule is heuristic; tuning ν in OC-SVM provides a principled control of the expected outlier fraction.

5 Conclusion

The pipeline leverages three cores: optimization dynamics (SGD→momentum→Adam), spectral graph embeddings (Laplacian eigenmaps), and kernel one-class SVMs. Each module not only flagged trial 419—showcasing distinct neural artifacts—but also provides complementary perspectives on anomaly scoring. Future work could incorporate convolutional/recurrent architectures, diffusion-map embeddings, and transformer-based detectors, as well as semi-supervised regression on stimulus labels.

References

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.